

PURCHASE DESCRIPTION

Replacement of Sewer Pumps and Controls at Cottage Grove Sewage Plant, Lane County Oregon.

1. SCOPE:

This Purchase Description covers the replacement of two Vertical Turbine sewer effluent pumps and a duplex pump controller for the Cottage Grove Sewage Plant to ensure efficient and reliable operation of the facility.

2. BACKGROUND:

Cottage Grove Reservoir Sewage System Overview and Upgrade Plan

The Cottage Grove Reservoir, located in the South Willamette Valley, is a vital hub for recreation and professional activities. It features a Class A campground, parks, and an office for the US Army Corps of Engineers. These facilities, equipped with restrooms, sinks, and showers, generate a significant volume of sewage. Included in the facilities is a Class A Campground. A Class A Campground is one that has flush restrooms, showers, paved roads, sanitary disposal stations, designated tent or trailer spaces, refuse containers, and potable water.

The current sewage management process is as follows:

1. Effluent is created by septic systems at each facility. With a total of 4 facilities.
2. This effluent is pumped to a large collection pond with a capacity of approximately 2 million gallons.
3. From the pond, the effluent is gravity-fed into a sewage treatment building.
4. Inside the building, the effluent is treated to meet the standards for discharge via irrigation into an effluent field.
5. The pumps are located inside the building.

The entire facility is regulated and permitted by the Oregon State Department of Environmental Quality (DEQ).

3. REQUIREMENTS:

Pumps (2) & Motors (2) Characteristics

The two (2) new Vertical Turbine Pumps and motors shall meet or exceed the following specifications:

#	Feature	Specification
1	Power	(2) 5-7.5 HP 240 VAC Single Phase Electric Motors
2	Motor Style	WPI or TEFC (Totally Enclosed, Fan-Cooled)
3	Protection	Thermostat for overheat protection and space heaters for humidity control
4	Flow Rate	100-160 GPM
5	Stages (Bowls)	Multistage 5-9
6	Intake Strainers	304SS (Stainless Steel)
7	Corrosion Control	Glass enameled bowl assemblies, Municipal Epoxy coatings (ID/OD Head, ID/OD Column, OD Bowl)
8	Lubrication	Open line shaft, product lubricated
9	Performance Rating	"1U" level Hydraulic Institute Testing Standard Table 14.6.3.4
10	Riser Pipe	4" Diameter Effluent Riser Pipe, 75" length before coupling
11	Irrigation Line	Discharge is 605' in length with 2" pipe
12	Shaft	¾" Impeller/Drive Shaft, 28" above motor base
13	Dimensions	Motor base: 12.5"; Bowl Stages total length: 5' ½"; Total Riser length: 135 3/8" (11.29')
14	Flanges	8 bolt face flanges for Discharge. USACE will provide template upon request for baseplate bolt configuration

Controller and Floats

The new controller should be capable of controlling the above-listed pumps and include the following features:

#	Feature	Specification
1	Power	120 VAC Control Power, 240 VAC Pump Line Voltage
2	Capacity	Capable of controlling the 2 motors detailed above
3	Enclosure	Lockable, Nema 12 minimum rating, with an enclosure heater
4	Monitoring	Elapsed Hour meter for each individual pump
5	Auxiliary Power	120 VAC control circuit that powers a 120 Volt outlet for a .6-amp chlorinating diaphragm pump. Chlorinating pump activated when pumps run

6 Relays	(4) Form A (Normally Open) and (1) Form B (Normally Closed) auxiliary relays
7 Controls	Start/Stop/Reset buttons for each pump, Hand/Off/Auto settings for each pump, green "power on" lights as indicators
8 Auto Circuits	Float and low-pressure switch controlled. Auto circuit operates timer. Pressure switch shuts off pump at low pressure indication for pump protection.
9 Auto Functions	Lead/Lag rotation, high-level alarm (red light) with lag pump activation, low-level shutoff (amber light), programmable digital timer
10 Floats	Narrow band floats with 25 ft cords
11 Protection	Breaker protection for each motor starter and overload protection for each contactor

Spare Parts Required

Quantity	Description
1	Motor starter
2	Extra floats
1	Extra Form A (Normally Open) Auxiliary Contact

Documentation

- Mechanical drawings and specification sheets for the pumps and motors must be provided.
- A schematic diagram of the controller must be provided.

4. PURCHASE OF:

Cottage Grove two (2) Sewer Pumps and Controls as outlined in the requirements section above.

5. SHIPPING:

Deliver all listed materials to the Cottage Grove Sewer Plant approximately 5 miles from Cottage Grove Oregon. The facility does not have an address. Therefore, delivery must be coordinated with USACE WVP GM Supervisor Tim Stratis Timarion.Stratis@usace.army.mil (541) 349-3047(see map below). The Government shall not be responsible for the cost of shipping. Supplies shall be delivered between Monday through Thursday 0800 – 1500. Excluding holidays.

